

1. Introduction

A **Payment Gateway system** is a core financial infrastructure component that enables businesses to accept and process digital payments from customers securely and reliably. Its primary goal is to facilitate the transfer of money from a customer to a merchant in exchange for goods or services.

At a high level, the system acts as an intermediary between multiple parties involved in a payment transaction, ensuring that payment data is securely captured, validated, authorized, and eventually settled.

Scope of the System

We are designing a full end-to-end payment lifecycle, which includes:

- Capturing payment details (e.g., card information) from the customer
- Securely transmitting payment data for authorization
- Communicating with external financial networks (banks and card networks)
- Handling success/failure responses
- Recording transaction states
- Ensuring eventual fund settlement to the merchant

Key Idea

The system shares strong similarities with a banking system in terms of:

- Transaction consistency
- Ledger maintenance
- Fraud detection and risk handling
- High reliability and fault tolerance requirements

However, a payment gateway adds additional complexity due to:

- Multiple external dependencies (banks, card networks, payment processors)
- Strict latency requirements
- Security constraints (PCI compliance, encryption, tokenization)
- Asynchronous settlement workflows

Domain Context

To properly design this system, a basic understanding of how **card payments work in the real world** is required, including:

- Card networks (Visa/Mastercard-like systems)

- Issuing banks and acquiring banks
- Authorization vs settlement flow
- Chargebacks and reversals